Support Vector Machine - Linear Classification

i) Start.

ii) Input: Training dataset (x_i, y_i) where $y_i \in \{-1, +1\}$

iii) Initialize weight vector w and bias b.

iv) Define objective:

- Maximize margin by minimizing: $\frac{1}{2}||w||^2$

  Subject to: $y_i(w \cdot x_i + b) \geq 1$

v) Solve the optimization problem using Lagrange multipliers or quadratic programming.

vi) Obtain optimal values of w and b.

vii) Construct decision function: $f(x) = w \cdot x + b$

viii) Classify a new sample:

- If $f(x) \geq 0$, assign class $+1$
- Otherwise, assign class -1

ix) End.